

HYPER-3

Which of three daily doses should go into phase III, and is any of them harming the people taking it?

Abstract

The study addresses which of three daily doses should go into phase III and whether any of them harms the people taking it. The effect of arm on change is identified via the backdoor route. Four hundred adults with elevated or stage-1 hypertension were randomized to standard of care or to one of three daily doses, stratified by age band. Seated systolic pressure was measured weekly, averaging readings inside windows for safety and primary endpoints. Each dose was compared with control by ANCOVA on the change from baseline, adjusting for baseline pressure and age band for pooled contrasts. Twelve harm contrasts were monitored against a posterior-probability boundary. For the pooled analysis, the 40 mg versus control contrast was -1.53 mmHg with a 90% WALD interval of -2.97 to -0.10 mmHg, the 20 mg versus control contrast was -4.80 mmHg with a 90% WALD interval of -6.17 to -3.44 mmHg, and the 10 mg versus control contrast was -5.27 mmHg with a 90% WALD interval of -6.63 to -3.91 mmHg. Within age bands, the 40 mg versus control contrast for age 51+ was 6.20 mmHg with a 90% WALD interval of 3.98 to 8.42 mmHg, and for age 25-35 was -5.26 mmHg with a 90% WALD interval of -8.17 to -2.36 mmHg. The analysis rests on the assumptions that dropout is ignorable given baseline and that the pooled contrast answers the question.

Reported quantities: 40 mg vs control, pooled; 20 mg vs control, pooled; 10 mg vs control, pooled; 40 mg vs control, age 51+

1. Introduction

This study addresses which of three daily doses should go into phase III, and whether any of them is harming the people taking it. The primary evaluation turns on the pooled contrast of 40 mg versus control. The target effect of arm on change is identified via the backdoor route. The evaluation of this contrast turns on a reference value of 0.00 mmHg. What counts as an answer depends directly on the location of the interval estimate relative to that value. An interval lying wholly on one side of 0.00 mmHg settles the question. An interval spanning 0.00 mmHg leaves the question unsettled and does not indicate the absence of an effect. The conclusions remain conditional throughout the analysis. Two unresolved assumptions carry the identification of the target contrast. First, the analysis assumes that dropout is ignorable given baseline. Second, it assumes that the pooled contrast answers the question. Both assumptions are maintained across the methods and qualify the interpretation of the contrast.

2. Methods

The target of evaluation is the effect of arm on change, which is identified via the backdoor route. To evaluate which of three daily doses should advance to phase III and whether any dose causes harm, 400 adults with elevated or stage-1 hypertension were randomized 2:1:1:1 to standard of care or to one of three daily doses, stratified by age band. The design estimates three contrasts against a shared control rather than evaluating one dose against another. Seated systolic pressure was measured weekly. Both endpoints average readings inside a window—weeks 5–8 for safety and weeks 9–12 for the primary endpoint—rather than using a single visit. Averaging several correlated readings roughly halves the variance of a unit's endpoint, allowing safety boundaries to operate within individual strata. Each dose was compared with control by ANCOVA on the change from baseline, adjusting for baseline pressure, and additionally adjusting for age band when pooling across bands. Randomization licenses the intent-to-treat contrast, and adjusting for pre-randomization covariates improves precision without forfeiting identification. Twelve harm contrasts, evaluating each dose against control overall and inside each age band, were monitored at scheduled safety reviews against a posterior-probability boundary with a 2 mmHg clinical margin. Pre-specifying the age bands as monitored contrasts enables detection of stratum-specific harms before the trial ends. The validity of these procedures rests on the assumptions that dropout is ignorable given baseline and that the pooled contrast answers the target question.

Assumptions this analysis rests on

Assumption	Statement	Status	Challenged by
dropout is ignorable given baseline	units who left the study differ from those who stayed only through what was measured before randomization	is assumed and has not been checked	a tipping-point analysis over plausible departures from ignorability
the pooled contrast answers the question	a single number over all age bands describes the effect the decision is about	is known to fail	heterogeneity across the pre-specified strata

Table 1. Standing assumptions

assumption Adherence was not adjusted for: it lies on the path from arm to outcome, and conditioning on it would break the randomization.

assumes *no_conditioning_on_post_randomization_variables* (satisfied): the analysis conditions on nothing measured after assignment; adherence is a mediator and dropout is a descendant of the outcome

assumption The pooled 40 mg contrast is reported alongside the per-band contrasts, never instead of them.

assumes *the_pooled_contrast_answers_the_question* (violated): a single number over all age bands describes the effect the decision is about

Assumption ledger

3. Results

The estimated effect of the 40 mg dose versus control, pooled across age bands, is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg). For the age 51 and older band, the estimated effect of 40 mg versus control is 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg).

40 MG VS CONTROL, POOLED	20 MG VS CONTROL, POOLED	10 MG VS CONTROL, POOLED	40 MG VS CONTROL, AGE 51+	40 MG VS CONTROL, AGE 25-35
-1.53 mmHg	-4.80 mmHg	-5.27 mmHg	6.20 mmHg	-5.26 mmHg
[-2.965, -0.1035] (90% WALD)	[-6.171, -3.437] (90% WALD)	[-6.631, -3.907] (90% WALD)	[3.985, 8.424] (90% WALD)	[-8.166, -2.363] (90% WALD)

4. Model checking

Model checking evaluates the fit of the analysis rather than the treatment effects. These diagnostics describe data behavior and sample retention across the study. Across all arms, 372 units contributed to the primary endpoint. The proportion of units still on study at the final week was 0.797. In the age strata, 100 units were randomized in band age 25 35. A total of 160 units were randomized in band age 36 50. In band age 51 plus, 140 units were randomized. The analysis leaves as an unresolved assumption that dropout is ignorable given baseline. A second unresolved assumption is that the pooled contrast answers the question. Diagnostic measures assess the consistency of the data with the fitted specification. These checks can only reduce confidence in the estimated numbers when discrepancies arise, never establish them. Meeting these diagnostics indicates the absence of detected fit failures rather than verification of the model assumptions.

UNITS STILL ON STUDY AT THE FINAL WEEK	UNITS CONTRIBUTING TO THE PRIMARY ENDPOINT	UNITS RANDOMIZED IN BAND AGE 25 35	UNITS RANDOMIZED IN BAND AGE 36 50	UNITS RANDOMIZED IN BAND AGE 51 PLUS
0.797	372	100	160	140

5. Discussion

The effect is identified via the backdoor route, allowing the estimated quantities to be interpreted as treatment effects rather than associations under the stated assumptions. For the pooled comparison of 10 mg against control, the estimated effect is -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg). This interval lies entirely on the favourable side of the decision threshold, settling the question in that direction on these data. For the pooled comparison of 20 mg against control, the estimated effect is -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg). This interval also falls entirely on the favourable side of the decision threshold. For the pooled comparison of 40 mg against control, the effect is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg), which likewise lies on the favourable side of the decision threshold. However, this pooled estimate for 40 mg should not be reported alone because it averages an age band that benefits with an age band that is harmed. In the age band 25 to 35, the effect of 40 mg compared with control is -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg), falling on the favourable side of the decision threshold. In the age band 51 and older, the effect of 40 mg compared with control is 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg), lying on the unfavourable side of the decision threshold. This contrast is what the pooled estimate conceals and what the safety boundary was monitored to detect. These evaluations remain conditional on the unresolved assumptions that dropout is ignorable given baseline and that the pooled contrast answers the question.

6. Conclusions

The determination of which daily dose should go into phase III and whether harm occurs cannot be answered by a single pooled estimate, conditional on the assumptions that dropout is ignorable given baseline and that the pooled contrast answers the question. The contrast for 40 mg versus control in the age 51+ band is 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg), which sits on the unfavourable side of the threshold. In contrast, 40 mg versus control in the age 25-35 band is -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg). When pooled across age bands, 40 mg versus control is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg). The pooled contrast for 20 mg versus control is -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg). The pooled contrast for 10 mg versus control is -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg). These other findings sit on the favourable side of the threshold. Any single summary averages harm with benefit across age strata. Because an average combines opposing effects, the disaggregated findings are the result. The pooled figure is not a substitute for the disaggregated stratum estimates. Each contrast is identified via the backdoor route. All findings hold under the unresolved assumptions that dropout is ignorable given baseline and that the pooled contrast answers the question.

7. Limitations

Two unresolved assumptions remain in the analysis. The first assumption is that dropout is ignorable given baseline. This assumption holds that units who left the study differ from those who stayed only through characteristics measured before randomization. This is assumed and has not been checked against the data. It licenses the analysis to proceed under baseline conditioning. A tipping-point analysis across plausible departures from ignorability would challenge this assumption. The second assumption is that the pooled contrast answers the question. This requires that a single number across all age bands describes the effect the decision is about. This assumption is known to fail. Heterogeneity across the pre-specified strata challenges reliance on the pooled contrast. Specifying each assumption clarifies the analytical steps that the data alone do not establish. It also identifies the precise conditions that would challenge each premise.

8. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
40 mg vs control, pooled	pooled_40	-1.53 mmHg (90% WALD -2.97 to -0.10 mmHg)	identify.ols (ANCOVA)
20 mg vs control, pooled	pooled_20	-4.80 mmHg (90% WALD -6.17 to -3.44 mmHg)	identify.ols (ANCOVA)
10 mg vs control, pooled	pooled_10	-5.27 mmHg (90% WALD -6.63 to -3.91 mmHg)	identify.ols (ANCOVA)
40 mg vs control, age 51+	oldest_40	6.20 mmHg (90% WALD 3.98 to 8.42 mmHg)	identify.ols (ANCOVA)
40 mg vs control, age 25-35	youngest_40	-5.26 mmHg (90% WALD -8.17 to -2.36 mmHg)	identify.ols (ANCOVA)
Units still on study at the final week	retention	0.797	—
Units contributing to the primary endpoint	analysed	372	—
Units randomized in band age 25 35	n_age_25_35	100	—
Units randomized in band age 36 50	n_age_36_50	160	—
Units randomized in band age 51 plus	n_age_51_plus	140	—

Table 2. Quantities and their sources

Field	Value
endpoint	mean change, weeks 9-12
interval_mass	90%
seed	20260821
world	nbs/case-studies/hypertension/hyper3.py
evidence hash	962d1a7970f3f2dfdc0297e7d9d6d16dd568a5441fc1cfaa7c44837d40f9c2e5
narration: abstract	gemini-3.5-flash-lite — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified

narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified
narration: conclusions	gemini-3.7-flash — verified
narration: limitations	gemini-3.7-flash — verified

Table 3. Run